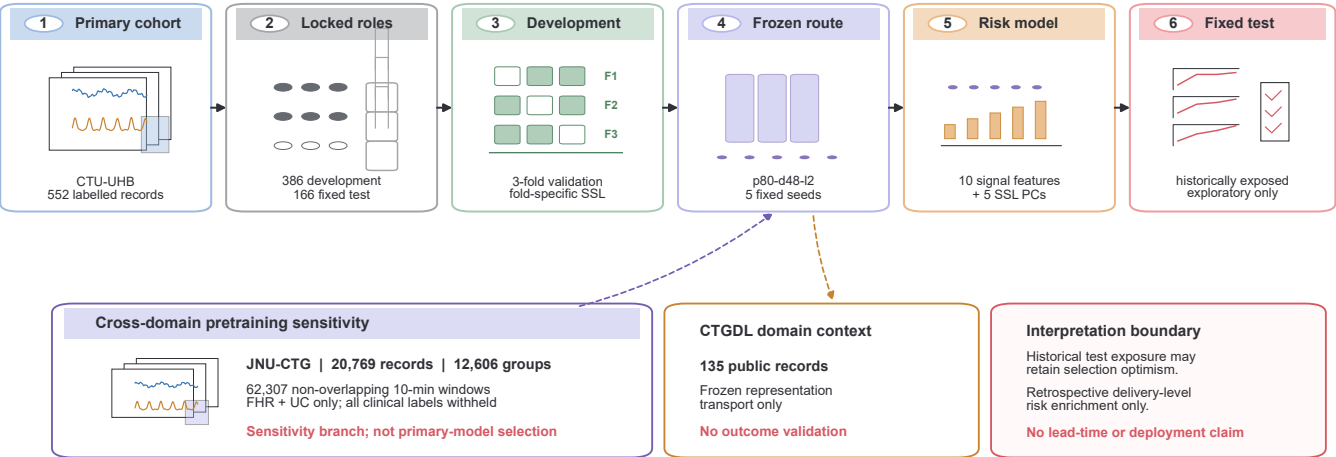


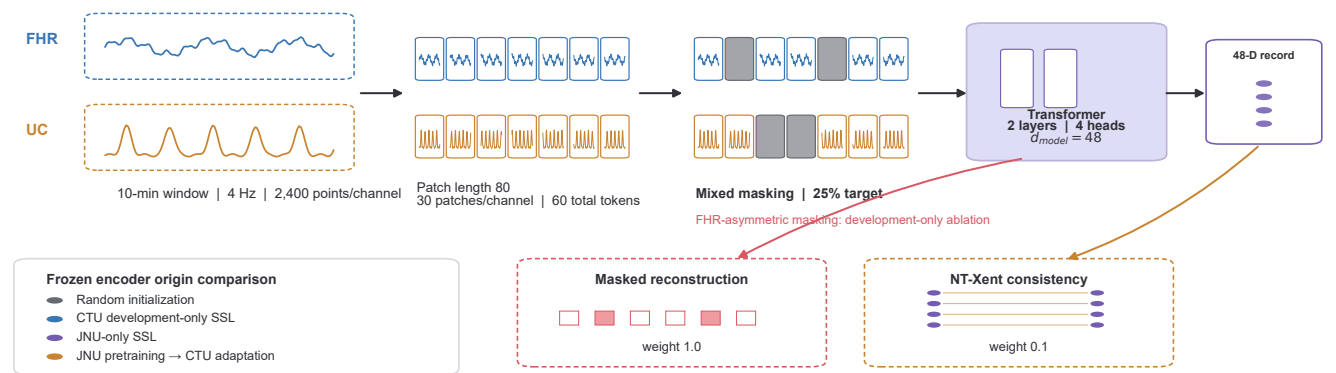
A Leakage-aware CTU evidence chain with bounded public pretraining sensitivity

CTU roles are locked before window generation; JNU labels are excluded from SSL and the fixed-test exposure is disclosed.



B Frozen PatchTST-style encoder and prespecified representation comparisons

Architecture and downstream dimensionality are unchanged; JNU pretraining is an additional sensitivity route.



C Public CTG scale and CTU outcome balance



D Analysis roles and claim boundaries

Question	Data role	Allowed claim
Representation learning	JNU + CTU development	labels withheld
Morphology probe	CTU expert events	representation task
Neonatal-risk enrichment	CTU development / fixed test	exploratory
External outcome validation	No harmonized outcome cohort	not performed

JNU sensitivity did not qualify for a primary risk-improvement claim; detailed results are supplementary.